

Graduate Level Elementary Particle Physics

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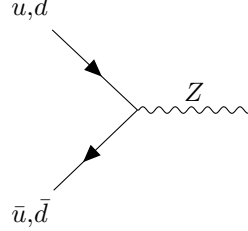
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*<https://github.com/yuriphestos/gradphestos-epp>

2 Task 2

2.1 Z Boson Production ($p\bar{p} \rightarrow Z \rightarrow l\bar{l}$) at the Tevatron

Calculate Z boson cross section in $p\bar{p}$ collision, and compare with observed rate ($p\bar{p} \rightarrow Z \rightarrow l\bar{l}$) at the Tevatron.



The total cross section of $p\bar{p} \rightarrow Z$ is given by

$$\sigma_{p\bar{p} \rightarrow Z} = \int_0^1 \int_0^1 dx_1 dx_2 f_{u,p}(x_1) f_{\bar{u},\bar{p}}(x_2) \sigma_{u\bar{u} \rightarrow Z}(x_1 x_2 s) + \int_0^1 \int_0^1 dx_1 dx_2 f_{d,p}(x_1) f_{\bar{d},\bar{p}}(x_2) \sigma_{d\bar{d} \rightarrow Z}(x_1 x_2 s)$$

One needs to find the cross section $\sigma_{q\bar{q} \rightarrow Z}$ using the following equation:

$$d\sigma = \frac{(2\pi)^4 |\mathcal{M}_{q\bar{q} \rightarrow Z}|^2}{4\sqrt{(p_q \cdot p_{\bar{q}})^2}} \delta^4(p_q + p_{\bar{q}} - p_Z) \frac{d^3 p_Z}{(2\pi)^3 2E_Z}$$

with the amplitude squared in terms of the total amplitude

$$|\mathcal{M}_{q\bar{q} \rightarrow Z}|^2 = \underbrace{\left(\frac{1}{3}\right)}_{\text{color}} \underbrace{\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)}_{\text{spin}} \sum_{\text{total}} |\mathcal{M}|^2$$

$$|\mathcal{M}_{Z \rightarrow q\bar{q}}|^2 = \frac{1}{3} \sum_{\text{total}} |\mathcal{M}|^2 \quad \Rightarrow \quad |\mathcal{M}_{q\bar{q} \rightarrow Z}|^2 = \frac{1}{4} |\mathcal{M}_{Z \rightarrow q\bar{q}}|^2$$

where $\hat{s} = (p_q + p_{\bar{q}})^2 = x_1 x_2 s$.

$$\begin{aligned} \sigma_{q\bar{q} \rightarrow Z} &= \frac{(2\pi)^4 |\mathcal{M}_{q\bar{q} \rightarrow Z}|^2}{4 \cdot \hat{s}/2} \frac{1}{(2\pi)^3 2E_Z} \delta(E_q + E_{\bar{q}} - E_Z) \\ &= \frac{\pi |\mathcal{M}_{q\bar{q} \rightarrow Z}|^2}{\hat{s}} \frac{\delta(E_q + E_{\bar{q}} - E_Z)}{E_q + E_{\bar{q}} + E_Z} \\ &= \frac{\pi |\mathcal{M}_{q\bar{q} \rightarrow Z}|^2}{\hat{s}} \delta((E_q + E_{\bar{q}})^2 - E_Z^2) \\ &= \frac{\pi |\mathcal{M}_{q\bar{q} \rightarrow Z}|^2}{\hat{s}} \delta(\hat{s} - m_Z^2) \\ &= \frac{\pi |\mathcal{M}_{Z \rightarrow q\bar{q}}|^2}{4\hat{s}} \delta(\hat{s} - m_Z^2) \end{aligned}$$

Using the decay width of Z we rewrite

$$\begin{aligned} \sigma_{q\bar{q} \rightarrow Z} &= \frac{4\pi^2 m_Z}{\hat{s}} \cdot \Gamma_{Z \rightarrow q\bar{q}} \cdot \delta(\hat{s} - m_Z^2) \\ &= \frac{4\pi^2}{m_Z} \cdot \Gamma_{Z \rightarrow q\bar{q}} \cdot \delta(\hat{s} - m_Z^2) \end{aligned}$$

Thus,

$$\begin{aligned}
\sigma_{p\bar{p} \rightarrow Z} &= \sum_{u,d} \int_0^1 \int_0^1 dx_1 dx_2 f_{q,p}(x_1) f_{\bar{q},\bar{p}}(x_2) \sigma_{q\bar{q} \rightarrow Z}(x_1 x_2 s) \\
&= \sum_{u,d} \int_0^1 \int_0^1 dx_1 dx_2 f_{q,p}(x_1) f_{\bar{q},\bar{p}}(x_2) \frac{4\pi^2}{m_Z} \Gamma_{Z \rightarrow q\bar{q}} \delta(x_1 x_2 s - m_Z^2) \\
&= \sum_{u,d} \int_{m_Z^2/s}^1 \frac{dx_1}{x_1 s} f_{q,p}(x_1) f_{\bar{q},\bar{p}}(m_Z^2/x_1 s) \frac{4\pi^2}{m_Z} \Gamma_{Z \rightarrow q\bar{q}} \\
&= \frac{4\pi^2}{m_Z s} \left[\Gamma_{Z \rightarrow u\bar{u}} \int_{m_Z^2/s}^1 \frac{dx_1}{x_1} f_{u,p}(x_1) f_{u,p}(m_Z^2/x_1 s) + \Gamma_{Z \rightarrow d\bar{d}} \int_{m_Z^2/s}^1 \frac{dx_1}{x_1} f_{d,p}(x_1) f_{d,p}(m_Z^2/x_1 s) \right]
\end{aligned}$$

By the charge conjugation symmetry of the proton and anti-proton, the anti-quark distribution in the anti-proton is identical to the quark distribution in the proton: $f_{\bar{q},\bar{p}}(x) = f_{q,p}(x)$. The valence quark PDFs for high x are approximated as:

$$f_u(x) = \frac{1}{0.38} x^{-0.52} (1-x)^{4.11}; \quad f_d(x) = \frac{1}{0.79} x^{-0.65} (1-x)^{5.08}$$

To evaluate the integral at the Tevatron center-of-mass energy $\sqrt{s} = 1.96$ TeV, we use the Standard Model parameters $m_Z \simeq 91.19$ GeV, $\Gamma_{Z \rightarrow u\bar{u}} \simeq 0.30$ GeV, and $\Gamma_{Z \rightarrow d\bar{d}} \simeq 0.40$ GeV. Including the leptonic branching ratio $\text{Br}(Z \rightarrow l\bar{l})$, the total Leading Order (LO) cross section can be evaluated:

$$\boxed{\sigma_{p\bar{p} \rightarrow l\bar{l}} = \sigma_{p\bar{p} \rightarrow Z} \times \text{Br}(Z \rightarrow l\bar{l}) \simeq 491 \text{ pb}}$$

Comparing this to the observed rates from the Tevatron (*e.g.*, as discussed by Dorigo), where the combined e and μ cross sections are roughly $267 + 246 = 513$ pb:

$$\frac{\sigma_{p\bar{p} \rightarrow l\bar{l}}}{\sigma_{\text{obs}}} = \frac{491 \text{ pb}}{513 \text{ pb}} \simeq 0.96$$

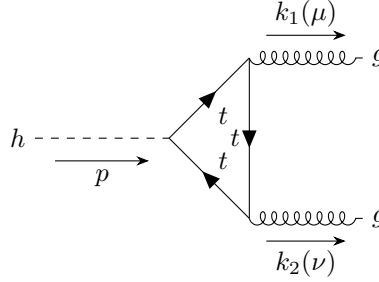
This tree-level calculation recovers $\sim 96\%$ of the observed experimental cross section. The slight underestimation is expected, as our calculation does not include Next-to-Leading Order (NLO) QCD corrections (the so-called K -factor) which typically enhance hadronic cross sections.

(see `task-2-1.m`)

2.2 Higgs Decay to Digluons ($h \rightarrow gg$) and Coupling

The coupling of the SM to gluons is induced by the quantum loops of the top quark. Let us take a crude approximation of very heavy top quark, and assume that the coupling is given by $\mathcal{L}_{hgg} = \frac{\alpha_s}{16\pi v} \times c_{hgg} \times h \times G_{\mu\nu}^a G_{\mu\nu}^a$, where $G_{\mu\nu}^a$ is the gluon field strength, $a = 1, \dots, 8$, and strong coupling constant $\alpha_s \simeq 0.11$ at relevant energies. Calculate the $h \rightarrow gg$ decay width as function of c_{hgg}^2 . Using your answer, the total SM Higgs width, and known answer for the $h \rightarrow gg$ branching ratio (Higgs review in PDG), extract the value of c_{hgg}^2 . Calculated value of the c_{hgg} can be found in *e.g.* hep-ph/9504378.

($m_t \rightarrow \infty$)



The effective $\mathcal{L}$ is given by

$$\mathcal{L}_{hgg} = \frac{\alpha_s}{16\pi v} \times c_{hgg} \times h \times G_{\mu\nu}^a G_{\mu\nu}^a$$

where $G_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + g_s f^{abc} A_\mu^b A_\nu^c$.

$$\begin{aligned} G_{\mu\nu}^a G_{\mu\nu}^a &= (\partial_\mu A_\nu^a - \partial_\nu A_\mu^a + g_s f^{abc} A_\mu^b A_\nu^c)(\partial_\mu A_\nu^a - \partial_\nu A_\mu^a + g_s f^{abc} A_\mu^b A_\nu^c) \\ &= 2(\partial_\mu A_\nu^a \partial_\mu A_\nu^a - \partial_\mu A_\nu^a \partial_\nu A_\mu^a) + 2g_s f^{abc}(\partial_\mu A_\nu^a - \partial_\nu A_\mu^a)A_\mu^b A_\nu^c + g_s^2 f^{abe} f^{cde} A_\mu^a A_\nu^b A_\mu^c A_\nu^d \end{aligned}$$

Only the first term contributes to $h - gg$ vertex so we are left with

$$G_{\mu\nu}^a G_{\mu\nu}^a = 2(\partial_\mu A_\nu^a \partial_\mu A_\nu^a - \partial_\mu A_\nu^a \partial_\nu A_\mu^a) = 2(g_{\mu\nu} k_1 \cdot k_2 - k_{1,\nu} k_{2,\mu})$$

Since we have two final state gluons, an additional combinatoric factor of 2 appears from the momentum space expansion of $\langle h | \partial_\mu A_\nu^a \partial_\mu A_\nu^a | g^b(k_1) g^c(k_2) \rangle$. Including the color delta function δ^{bc} , the matrix element is:

$$\mathcal{M} = \frac{\alpha_s}{16\pi v} c_{hgg} \times 4 \times \delta^{bc} [g_{\mu\nu}(k_1 \cdot k_2) - k_{1,\nu} k_{2,\mu}] \epsilon_{1,\mu}^* \epsilon_{2,\nu}^*$$

To find the squared amplitude, we sum over the 8 gluon colors ($\delta^{bc} \delta^{bc} = 8$) and sum over the transverse gluon polarizations using $\sum \epsilon_\mu \epsilon_{\mu'}^* \rightarrow -g_{\mu\mu'}$:

$$\sum_{\text{colors, pols}} |\mathcal{M}|^2 = \left(\frac{\alpha_s c_{hgg}}{4\pi v} \right)^2 \times 8 \times 2(k_1 \cdot k_2)^2$$

Using the kinematic relation $2(k_1 \cdot k_2) = m_h^2$, this becomes:

$$\sum_{\text{colors, pols}} |\mathcal{M}|^2 = \frac{\alpha_s^2 c_{hgg}^2}{16\pi^2 v^2} \times 16 \times \left(\frac{m_h^2}{2} \right)^2 = \frac{\alpha_s^2 c_{hgg}^2 m_h^4}{4\pi^2 v^2}$$

To find the decay width, we integrate over the 2-body phase space. Because the final state consists of two identical massless gluons, we must include a symmetry factor of $S = 1/2!$:

$$\begin{aligned}\Gamma_{h \rightarrow gg} &= \frac{1}{2!} \frac{1}{2m_h} \int \frac{d^3k_1}{(2\pi)^3 2E_1} \frac{d^3k_2}{(2\pi)^3 2E_2} |\mathcal{M}|^2 (2\pi)^4 \delta^4(p_h - k_1 - k_2) \\ &= \frac{\alpha_s^2 c_{hgg}^2 m_h^3}{256\pi^4 v^2} \int \frac{k_2 dk_2 d\Omega}{m_h - k_2} \frac{\delta(k_2 - m_h/2)}{2} = \boxed{\frac{\alpha_s^2 c_{hgg}^2 m_h^3}{128\pi^3 v^2}}\end{aligned}$$

Using the PDG values $\Gamma_h = 4.07 \times 10^{-3}$ GeV and $\text{Br}(h \rightarrow gg) = 8.6\%$:

$$\Gamma_{h \rightarrow gg} = 3.5 \times 10^{-4} \text{ GeV} = \frac{\alpha_s^2 c_{hgg}^2 m_h^3}{128\pi^3 v^2} \quad \Rightarrow \quad \boxed{c_{hgg} \simeq 1.88}$$

Comparison with hep-ph/9504378:

The effective Lagrangian derived from integrating out the heavy top quark loop is:

$$\mathcal{L}_{hgg} = \frac{\alpha_s}{12\pi} \left(\sqrt{2} G_F \right)^{1/2} \left(1 + \frac{11}{4} \frac{\alpha_s}{\pi} \right) G_{\mu\nu}^a G_{\mu\nu}^a h$$

Equating the coefficients of $G_{\mu\nu}^a G_{\mu\nu}^a h$ from this expression and our ansatz, and using $v = (\sqrt{2} G_F)^{-1/2} \simeq 246$ GeV:

$$\frac{\alpha_s}{16\pi v} c_{hgg} = \frac{\alpha_s}{12\pi} \left(\sqrt{2} G_F \right)^{1/2} \left(1 + \frac{11}{4} \frac{\alpha_s}{\pi} \right)$$

$$c_{hgg} = \frac{4}{3} \left(1 + \frac{11}{4} \frac{\alpha_s}{\pi} \right) \simeq \boxed{1.46}$$

The empirical value ($c_{hgg} \simeq 1.88$) extracted from the PDG branching ratio is higher than the theoretical value (1.46). This is because the physical decay width $\Gamma_{h \rightarrow gg}$ includes massive Next-to-Leading Order (NLO) QCD corrections (a K -factor of $K \approx 1.6 - 1.7$). By fitting the fully QCD-corrected experimental width to a tree-level Leading Order (LO) formula, our extracted c_{hgg}^2 artificially absorbed this K -factor ($1.46 \times \sqrt{1.65} \approx 1.88$).

2.3 Higgs Production via Gluon Fusion at the LHC

Consider the resonant production of the Higgs boson at the LHC due to the dominant $gg \rightarrow h$ sub-process known as gluon fusion. Calculate the production cross section for 7, 8 and 13 TeV. (As in $q\bar{q} \rightarrow Z$ case, your parton level cross section will have $\Gamma_{hgg}/m_h \times \delta(\hat{s} - m_h^2)$. Pay attention to gluon spin and color averages.) Compare with the known predictions/observations of the Higgs production at 7, 8 and 13 TeV. You should have an order-of-magnitude agreement despite a simplified nature of our treatment.

Because the LHC is a pp collider, the total cross section of $pp \rightarrow h$ via gluon fusion is given by:

$$\sigma_{pp \rightarrow h} = \int_0^1 \int_0^1 dx_1 dx_2 f_g(x_1) f_g(x_2) \sigma_{gg \rightarrow h}(x_1 x_2 s)$$

Recall from Section 2.2 that the decay width to two identical gluons is:

$$\Gamma_{h \rightarrow gg} = \frac{1}{32\pi m_h} \sum |\mathcal{M}|^2 \implies \sum |\mathcal{M}|^2 = 32\pi m_h \Gamma_{h \rightarrow gg}$$

The partonic cross section incorporates averages over the initial gluon colors (8 each) and transverse polarizations (2 each):

$$\begin{aligned} \sigma_{gg \rightarrow h}(x_1 x_2 s) &= \underbrace{\left(\frac{1}{8}\right) \left(\frac{1}{8}\right)}_{\text{color}} \underbrace{\left(\frac{1}{2}\right) \left(\frac{1}{2}\right)}_{\text{polarization}} \frac{\pi}{\hat{s}} \sum |\mathcal{M}|^2 \delta(\hat{s} - m_h^2) \\ &= \frac{1}{256} \frac{\pi}{m_h^2} (32\pi m_h \Gamma_{h \rightarrow gg}) \delta(x_1 x_2 s - m_h^2) \\ &= \frac{\pi^2}{8m_h} \cdot \Gamma_{h \rightarrow gg} \cdot \delta(x_1 x_2 s - m_h^2) \end{aligned}$$

where we evaluated $1/\hat{s}$ at the resonance $\hat{s} = (p_g + p_g)^2 = m_h^2 = x_1 x_2 s$. Thus, integrating over the momentum fractions yields:

$$\sigma_{pp \rightarrow h} = \int_0^1 \int_0^1 dx_1 dx_2 f_g(x_1) f_g(x_2) \frac{\pi^2}{8m_h} \Gamma_{h \rightarrow gg} \delta(x_1 x_2 s - m_h^2)$$

Evaluating the delta function sets $x_2 = m_h^2/(x_1 s)$, which restricts the lower bound of x_1 to m_h^2/s :

$$\sigma_{pp \rightarrow h} = \frac{\pi^2}{8m_h s} \Gamma_{h \rightarrow gg} \int_{m_h^2/s}^1 \frac{dx_1}{x_1} f_g(x_1) f_g\left(\frac{m_h^2}{x_1 s}\right)$$

We use the high- x gluon PDF approximation $f_g(x) = 4.71^{-1} x^{-1.79} (1-x)^{5.81}$, alongside the PDG values $\Gamma_{h \rightarrow gg} = 3.5 \times 10^{-4}$ GeV and $m_h = 125$ GeV.

$(\sqrt{s} = 7 \text{ TeV})$	$\sigma_{pp \rightarrow h} \simeq 7.34 \text{ pb}$
$(\sqrt{s} = 8 \text{ TeV})$	$\sigma_{pp \rightarrow h} \simeq 9.77 \text{ pb}$
$(\sqrt{s} = 13 \text{ TeV})$	$\sigma_{pp \rightarrow h} \simeq 26.7 \text{ pb}$

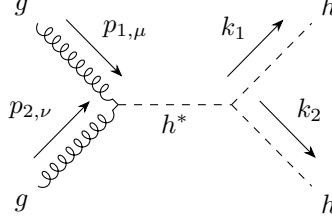
Comparison with known production (PDG ggF):

$$\begin{aligned}(\sqrt{s} = 7 \text{ TeV}) & \quad \sigma_{ggF} \simeq 16.9 \text{ pb} \\(\sqrt{s} = 8 \text{ TeV}) & \quad \sigma_{ggF} \simeq 21.4 \text{ pb} \\(\sqrt{s} = 13 \text{ TeV}) & \quad \sigma_{ggF} \simeq 48.6 \text{ pb}\end{aligned}$$

Our Leading Order (LO) calculation successfully achieves the expected order-of-magnitude agreement with the observed data. The observed cross sections are roughly a factor of ~ 2 higher due to massive Next-to-Leading Order (NLO/NNLO) QCD corrections (K -factors) that drastically enhance gluon fusion rates at the LHC.

2.4 Di-Higgs Production and Luminosity Sensitivity

Using the SM value of c_{hgg} (hep-ph/9504378), estimate the di-Higgs production at the LHC via $gg \rightarrow h^* \rightarrow hh$. (In practice, there are other contributions, *e.g.*, when both Higgs bosons are attached to the top loop, but let us neglect this subtlety for now. Use simplified pdfs.) What integrated luminosity should LHC experiments reach to start being sensitive to the hh channel?



hgg vertex (including permutations):

$$\frac{\alpha_s}{4\pi v} c_{hgg} (g_{\mu\nu} p_1 \cdot p_2 - p_{1,\nu} p_{2,\mu})$$

h^* propagator:

$$\frac{1}{(p_1 + p_2)^2 - m_h^2}$$

Triple Higgs vertex:

$$-\frac{3i}{2} \frac{g_W m_h^2}{m_W} = -3i \frac{m_h^2}{v}$$

Matrix Element:

$$\mathcal{M} = -\frac{\alpha_s}{4\pi v} c_{hgg} (g_{\mu\nu} p_1 \cdot p_2 - p_{1,\nu} p_{2,\mu}) \varepsilon_\mu(p_1) \varepsilon_\nu(p_2) \frac{1}{(p_1 + p_2)^2 - m_h^2} \frac{3m_h^2}{v}$$

To find the spin- and color-averaged squared amplitude, we average over the initial gluons and multiply by the color trace of the vertex ($\delta^{ab}\delta^{ab} = 8$):

$$\langle |\mathcal{M}|^2 \rangle = \underbrace{\left(\frac{1}{64} \right)}_{\text{color avg}} \times \underbrace{\left(\frac{1}{4} \right)}_{\text{spin avg}} \times 8 \times |\mathcal{M}|^2 = \frac{9\alpha_s^2 c_{hgg}^2 m_h^4}{512\pi^2 v^4} \frac{2(p_1 \cdot p_2)^2}{[(p_1 + p_2)^2 - m_h^2]^2} = \frac{9\alpha_s^2 c_{hgg}^2 m_h^4}{1024\pi^2 v^4} \frac{\hat{s}^2}{(\hat{s} - m_h^2)^2}$$

To compute the cross section, we include the standard flux factor ($1/2\hat{s}$), the 2-body phase space, and a symmetry factor of $1/2!$ because the final state consists of two identical Higgs bosons:

$$\begin{aligned} \sigma_{gghh} &= \frac{1}{2\hat{s}} \frac{1}{2!} \int d\Pi_2 \langle |\mathcal{M}|^2 \rangle = \frac{1}{4\hat{s}} \left(\frac{1}{4\pi} \frac{\sqrt{\hat{s}/4 - m_h^2}}{\sqrt{\hat{s}}} \right) \langle |\mathcal{M}|^2 \rangle \\ &= \frac{1}{16\pi\hat{s}^2} \sqrt{\hat{s}^2/4 - \hat{s}m_h^2} \left(\frac{9\alpha_s^2 c_{hgg}^2 m_h^4}{1024\pi^2 v^4} \frac{\hat{s}^2}{(\hat{s} - m_h^2)^2} \right) \\ &= \frac{9\alpha_s^2 c_{hgg}^2 m_h^4}{16384\pi^3 v^4} \frac{\sqrt{\hat{s}^2/4 - \hat{s}m_h^2}}{(\hat{s} - m_h^2)^2} \\ &= N \cdot \frac{\sqrt{\hat{s}^2/4 - \hat{s}m_h^2}}{(\hat{s} - m_h^2)^2} \end{aligned}$$

hh production using PDFs at the LHC ($\hat{s} \rightarrow x_1 x_2 s$):

$$\sigma_{pp \rightarrow hh} = N \int_0^1 \int_0^1 dx_1 dx_2 f_g(x_1) f_g(x_2) \frac{\sqrt{(x_1 x_2 s)^2/4 - m_h^2 x_1 x_2 s}}{(x_1 x_2 s - m_h^2)^2}$$

To evaluate this efficiently, we change variables to $t_1 = x_1 x_2$ and $t_2 = x_1/x_2$, yielding $dx_1 dx_2 = dt_1 dt_2 / 2t_2$. Assuming the high- x approximation $f_g(x) \propto x^{-a}$ limits $(1-x)^n \rightarrow 1$, the integral becomes:

$$\begin{aligned} \sigma_{pp \rightarrow hh} &= \frac{N}{(4.71)^2} \int_0^1 \int_0^1 dx_1 dx_2 (x_1 x_2)^{-1.79} \frac{\sqrt{(x_1 x_2 s)^2/4 - m_h^2 x_1 x_2 s}}{(x_1 x_2 s - m_h^2)^2} \\ &= \frac{N}{2(4.71)^2} \int_{4m_h^2/s}^1 \int_{t_1}^{1/t_1} \frac{dt_1 dt_2}{t_2} (t_1)^{-1.79} \frac{\sqrt{(t_1 s)^2/4 - m_h^2 t_1 s}}{(t_1 s - m_h^2)^2} \end{aligned}$$

Note that the lower limit for t_1 is dictated by the kinematic threshold required to produce two on-shell Higgs bosons: $\hat{s} \geq 4m_h^2$.

Luminosity Sensitivity:

Using the theoretical SM value $c_{hgg} = 1.46$, numerical integration evaluates the total cross section at $\sqrt{s} = 13$ TeV to be:

$$\sigma_{pp \rightarrow hh} \simeq 7.00 \text{ fb}$$

To begin effectively probing this channel and conducting precision measurements, experiments generally require a minimum threshold of $\sim 10^3$ events. The integrated luminosity ($\mathcal{L}$) required to reach this sensitivity is:

$$\mathcal{L} = \frac{\text{Number of Events}}{\sigma_{pp \rightarrow hh}} = \frac{1000}{7.00 \text{ fb}} \simeq \boxed{143 \text{ fb}^{-1}}$$